

Review

Deep learning for non-contact machinery fault diagnosis: A review

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ABSTRACT

In industrial production settings, mechanical equipment may suffer from wear, cracking, and fracture due to harsh conditions such as variable loads, high impacts, and strong disturbances, leading to performance degradation, production halts, and even catastrophic accidents. With the rapid advancement of deep learning, neural network-based approaches have emerged as the dominant paradigm for intelligent fault diagnosis, enabling automatic hierarchical feature extraction and complex pattern recognition from multi-source non-contact sensing data. Traditional contact based diagnostic methods, despite their high precision, are constrained by deployment limitations in extreme environments. Non-contact fault diagnosis technology, centered on acoustic sensors, infrared thermal imagers, and laser Doppler vibrometers, fundamentally eliminates operational risks associated with physical connections while providing innovative solutions for equipment monitoring in hazardous conditions. This review first introduces the basic background and current development status of non-contact fault diagnosis. It then classifies and discusses non-contact fault diagnosis methods, including acoustic emission, vibration and thermal imaging. Finally, existing challenges in current methods and potential future development directions are analyzed.

1. Introduction

The rapid evolution of industrial automation and intelligent manufacturing has intensified reliability and safety demands on mechanical equipment (Jiang et al., 2025; Zhu et al., 2025). To ensure the stable operation of the equipment, timely inspection and maintenance are of vital importance. Fault diagnosis technology, as a pivotal means of preventing unexpected downtime and ensuring production continuity, has attracted considerable research interest in recent years (Qin et al., 2026; Tian et al., 2026; Wu et al., 2025). Through continuous monitoring of rotating machinery conditions and real-time analysis of anomalous signals, this technology facilitates accurate detection of incipient failures, equipping maintenance personnel with actionable insights for precision maintenance (Jiang et al., 2023). As a result, the operational reliability of equipment is significantly enhanced, and the risk of unplanned downtime are effectively mitigated.

It is worth noting that the root causes of mechanical failures extend beyond natural degradation processes, such as fatigue, wear, and material aging under variable loads and harsh environments. The increasing convergence of operational technology and information technology in modern industrial ecosystems has exposed rotating machinery to cyber-physical threats, wherein adversaries can intentionally induce faults by exploiting vulnerabilities in networked control systems. For example,

malicious manipulation of variable frequency drives may inject spurious speed or torque commands, forcing bearings to operate at critical resonant frequencies or under abnormal load spectra, thereby accelerating contact fatigue, lubrication deterioration, and spalling while altering the characteristic fault frequencies that conventional diagnosis models rely upon. Such attack-induced degradation trajectories deviate from established physics-based wear patterns, posing significant challenges to traditional fault diagnosis systems that assume failures arise solely from stochastic mechanical processes.

At present, contact-based sensors constitute the primary means of data acquisition in rotating machinery health monitoring. Their direct physical contact with equipment facilitates high-fidelity signal capture, supporting precise and reliable fault diagnosis that has led to broad industrial adoption (Carden & Fanning, 2004; Chen et al., 2024b, 2023). Although conventional contact-based diagnosis methodologies have achieved considerable technical maturity, their practical deployment confronts several critical challenges. First, sensor installation procedures are inherently complex and may disrupt normal equipment operation. Moreover, implementation becomes prohibitively difficult under extreme conditions such as high temperatures, high pressures, or toxic environments, thereby constraining their applicability. Second, as modern industrial equipment grows increasingly sophisticated, the limitations of contact-based methods become more pronounced.

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Table 1
Comparison between contact-based and non-contact fault diagnosis.

Attribute	Contact	Non-contact
Contact way	Sensor directly contacts the surface of the measured object	Sensor maintains a certain distance from the measured object with no physical contact
Signal acquisition	Transmits vibration, temperature, etc. through mechanical coupling	Remotely senses through electromagnetic, optical, acoustic field signals
Installation	Requires shutdown for installation or embedding inside equipment	Online, non-stop detection possible
Typical sensors	Accelerometers, strain gauges, thermocouples, contact microphones	Infrared thermal imagers, laser vibrometers, microphone arrays, cameras, eddy current sensors
Environmental interference susceptibility	Low, but affected by oil contamination, humidity, surface roughness	High, affected by lighting, temperature, airflow, electromagnetic interference, etc.
Device intrusiveness	High, may affect equipment operation or require shutdown for installation	Low, no shutdown required, does not affect equipment operation
Real-time monitoring capability	Strong, suitable for continuous online monitoring	Strong, also suitable for continuous online monitoring with no wear
Data processing complexity	Lower, relatively clear signal structure	Higher, requires handling low signal-to-noise ratio and multi-source heterogeneous data
Requirement for equipment surface	High (needs to be clean, flat, accessible)	Low (can work through windows, obstacles, or at a distance)
Safety	Risks of electrical, thermal, mechanical contact	High, no physical contact risk

Specifically, long-distance signal transmission induces attenuation in the vibration response of mechanical connection components. This effect, compounded by intense environmental noise interference, severely compromises the accuracy and reliability of monitoring signals (Chen et al., 2022; Sun et al., 2023a; Tian et al., 2024). Additionally, vibration signal acquisition is susceptible to surface conditions, including oil contamination, moisture levels, and irregular geometric morphology, which significantly degrades signal quality. In light of these technical bottlenecks, the development of non-contact fault diagnosis methods holds substantial engineering value. This technology effectively circumvents the inherent deficiencies of traditional contact-based monitoring and offers innovative solutions for equipment condition monitoring in extreme operating conditions or hazardous environments. Table 1 summarizes the differences between contact and non-contact fault diagnosis.

Non-contact fault diagnosis technology, which centers on monitoring methods requiring no physical contact with equipment surfaces such as infrared thermography, ultrasonic detection, and acoustic monitoring, enables real-time assessment of equipment operating conditions while fundamentally eliminating the operational risks associated with physical connections inherent in traditional contact-based inspection. By integrating advanced sensing, intelligent signal processing, and big data analytics, this technology captures operational anomalies in real time, provides early warnings of potential failures, and significantly enhances equipment reliability and maintenance efficiency. Leveraging its non-contact advantages, this technology has found extensive applications in critical sectors including manufacturing, energy, transportation, and aerospace. Currently, mainstream non-contact diagnosis methods primarily encompass acoustic detection technology, infrared thermography, and non-contact vibration monitoring. These approaches achieve precise evaluation of equipment status through remote, non-destructive monitoring while avoiding interference with equipment operation inherent in traditional contact-based methods, thereby offering innovative solutions for intelligent operation and maintenance of industrial equipment. An overview of non-contact fault diagnosis is illustrated in Fig. 1.

To promote the advancement of non-contact fault diagnosis technology, this review provides a systematic and timely synthesis of existing

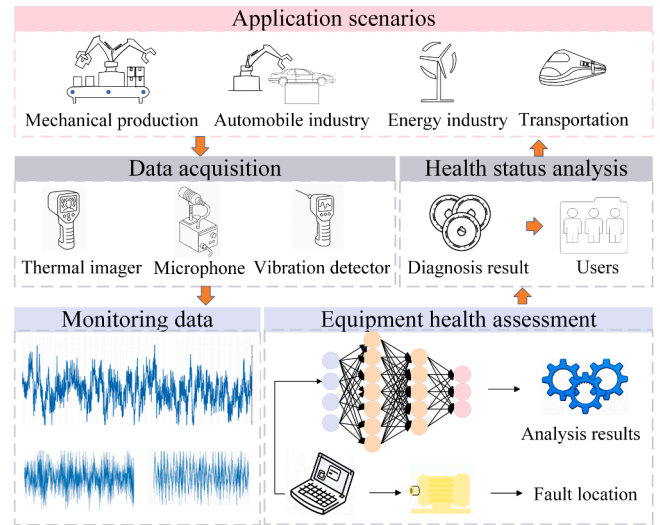


Fig. 1. Overview of non-contact fault diagnosis.

research across multiple dimensions, encompassing sensor modalities, diagnostic methodologies, and future directions. The principal contributions of this article are as follows:

1. A systematic classification framework is established from complementary perspectives of sensor modalities and diagnostic methodologies, enabling systematic organization of non-contact sensing techniques and intelligent analysis approaches.
2. Fundamental challenges are identified, including low signal-to-noise ratio (SNR), poor cross-domain generalization, feature redundancy, and model interpretability deficits that limit field reliability in complex industrial environments.
3. A forward-looking roadmap is proposed, advancing four transformative directions, i.e., physics-informed data-driven methods, multi-sensor fusion, expert knowledge integration, and large language

model adaptation, to guide the field toward knowledge-guided intelligent diagnosis.

The rest of this article is organized as follows. Section 2 provides an overview of non-contact sensing technologies, including acoustic sensors, infrared thermal imagers, laser Doppler vibrometers, high-speed cameras, electrostatic sensors, Ultra-Wideband (UWB) radar, millimeter-wave radar, and laser displacement sensors. Section 3 systematically reviews non-contact fault diagnosis methods, covering time-frequency analysis and signal decomposition techniques, statistical methods, neural network-based approaches with emphasis on attention mechanisms and transfer learning, and feature fusion strategies. Section 4 discusses the current challenges and future research directions, including physics-informed data-driven methods, multi-sensor fusion, integration of expert knowledge with deep learning, and large language models for fault diagnosis. Finally, Section 5 concludes the paper.

2. Overview of non-contact sensing technologies

Commonly used non-contact sensors primarily include acoustic sensors (microphones), infrared thermal imagers, and non-contact vibration sensors, etc. Each category is detailed in the following subsections.

2.1. Acoustic sensors

Non-contact fault diagnosis based on acoustic sensors leverages acoustic signals for machinery health monitoring (Chen et al., 2019; Hao et al., 2007; He et al., 2022; Hou et al., 2022, 2019; Liu et al., 2023; Wang et al., 2020; Wu & Meng, 2006; Xiao et al., 2024; Zhang et al., 2022a). By analyzing anomalous acoustic features extracted from sound data collected during equipment operation, this approach enables precise detection of incipient mechanical faults.

Compared with conventional contact-based methods requiring embedded or surface-mounted sensors, this technique completely eliminates added mass, electrical interference, and safety hazards associated with mechanical coupling. It is particularly suitable for scenarios involving high-speed rotation, extreme temperatures and pressures, or enclosed cavities where close-range sensor deployment is impractical. The typical implementation involves deploying miniature acoustic sensors at critical nodes within the sound field surrounding the target equipment. Beamforming and near-field acoustic holography are employed to enhance SNR, followed by spectral subtraction, cepstral analysis, or time-frequency sparse decomposition to isolate weak transient features indicative of faults.

Acoustic sensors offer the advantages of low cost and easy deployment. Combined with well-established signal processing frameworks (Bao et al., 2021; Jung et al., 2021a; Li et al., 2024a), they have been widely adopted in complex industrial scenarios such as wind turbine gearboxes and high-speed train bearings. The introduction of pattern recognition algorithms, including attention mechanisms and self-supervised contrastive learning, has further enhanced the sensitivity, robustness, and cross-condition generalization capability of acoustic-based fault diagnosis, providing a feasible technical pathway for non-contact intelligent operation and maintenance.

2.2. Infrared thermal imager

Infrared thermal imaging technology provides another important avenue for non-contact fault diagnosis. By capturing long-wave infrared radiation in the 7.5–14 μm spectral band emitted from equipment surfaces, thermal imagers can generate high-resolution temperature distribution maps within milliseconds, enabling non-contact quantitative analysis of early fault indicators such as localized overheating and abnormal cooling.

To ensure measurement accuracy, the thermal imager must be calibrated against blackbody radiation based on the target emissivity, while

atmospheric attenuation, background radiation, and reflected interference in the optical path should be incorporated into the radiative transfer model for correction. During on-site deployment, the thermal imager is typically mounted on a rigid bracket at a distance of 0.5–3 m from the target, with the optical axis centered on equipment hotspots using a laser pointer. Meanwhile, infrared transmissive windows or nitrogen purging are employed to mitigate interference from steam and dust. After non-uniformity correction, spatial registration, and temperature drift compensation of raw thermal images, the original thermal image can be used to extract the spatio-temporal features of thermal anomalies through deep neural networks, enabling early warning of faults such as bearing lubrication deterioration and motor winding short circuit (Guo et al., 2024; Hong & Kim, 2014; Jakubek et al., 2022; Janssens et al., 2015; Jia et al., 2019; Li et al., 2024b; Lim et al., 2014; Liu et al., 2017; Morales-Perez et al., 2019; Wang et al., 2018).

2.3. Laser doppler vibrometer

The Laser Doppler Vibrometer (LDV) operates on the Doppler effect principle and represents a high-precision, non-contact measurement instrument specifically designed for characterizing the dynamic vibration behavior of objects. This device can achieve nanometer-accuracy vibration displacement measurements by detecting the Doppler frequency shift of laser beams reflected from the target surface. During the data collection process, representative measurement points are first selected to calibrate the measuring instrument. Then, the vibration meter is fixed on a vibration-proof platform or tripod, and ensure that the laser beam is incident perpendicularly onto the target surface. To achieve optimal SNR, the target surface should have good optical reflection properties. In case of necessity, special reflective patches can be attached to enhance the signal quality.

With sub-micrometer measurement resolution and wide-frequency response characteristics, LDV has become an essential tool in mechanical dynamics analysis, structural modal testing, and rotating machinery fault diagnosis. Its non-contact measurement mechanism can eliminate dynamic interference from added mass on the test object, making it particularly suitable for high-precision applications or lightweight structures, and effectively preventing dynamic characteristic distortion caused by loading effects inherent to traditional contact-based sensors.

2.4. High-speed camera

High-speed cameras achieve the visualization recording of transient processes through sub-millisecond temporal sampling rates, with typical frame rates spanning the $10^2 - 10^4$ fps range. Ultra-high-speed models such as the Phantom VEO series can even surpass the 10^4 fps threshold. This high temporal resolution capability allows them to resolve extremely short-duration physical events, including high-speed motion trajectories, transient deformations, and shock wave propagation, thereby providing direct visual evidence for kinetic parameter extraction and failure mechanism analysis.

For fault diagnosis data acquisition based on high-speed cameras, the equipment should be deployed in the near-field region of the bearing, with the lens precisely aligned toward critical moving components such as rolling elements, inner race, and outer race. The camera angle and position should be dynamically adjusted according to the monitoring target to optimize imaging quality.

2.5. Electrostatic sensor

Electrostatic sensors achieve non-contact monitoring of rotating machinery vibration through electrostatic field coupling mechanisms, with the core sensing unit employing high-sensitivity charge detection circuits featuring a detection limit of 0.1 pC. The signal generation mechanism involves shaft vibration modulating the spatiotemporal periodicity

of surface charge distribution, which excites nanoampere-level displacement currents on the electrode array. This weak signal is conditioned by high-gain transimpedance amplifiers (10^6 – 10^8 V/A) and narrowband filtering networks (passband 0.1–10 kHz), after which shaft orbit trajectories are extracted through signal reconstruction algorithms with positioning accuracy better than ± 50 μ m.

2.6. UWB radar and millimeter-wave radar

Both UWB radar and Millimeter Wave (mmWave) radar enable non-contact vibration monitoring without the need for retro-reflective targets or surface preparation, and both have been successfully applied to motor system fault diagnosis. However, they differ fundamentally in operating principles and performance characteristics. UWB radar transmits nanosecond-scale narrow pulses (pulse width < 1 ns) and achieves sub-millimeter displacement accuracy (± 0.1 mm). Its ultra-wide bandwidth (typically > 500 MHz) provides excellent range resolution, making it particularly suitable for capturing high-frequency vibration components (5–20 kHz) such as those generated by early bearing faults. In contrast, mmWave radar operates on frequency-modulated continuous-wave (FMCW) principles with a much higher carrier frequency (e.g., 77 GHz). The short wavelength (≈ 3.9 mm at 77 GHz) enables finer vibration resolution down to 0.01 mm. Moreover, mmWave radar systems often integrate multiple-input multiple-output (MIMO) antenna arrays, allowing 16-channel synchronous measurements and full three-dimensional vibration modal reconstruction of motor stators. In summary, UWB radar excels in detecting high-frequency transient vibrations with good range resolution, while mmWave radar offers superior displacement resolution and multi-channel spatial measurement capabilities, making it more suitable for detailed modal analysis.

2.7. Laser displacement sensor

The laser displacement sensor is a non-contact precision measurement device based on the optical triangulation principle, primarily employed for real-time monitoring of dynamic displacement in rotating machinery shaft systems. This sensor projects a parallel laser beam onto the measured shaft surface, with the reflected beam captured by a high-sensitivity photodetector. By analyzing the positional offset of the reflected light spot on the detector, dynamic displacement measurements with micrometer-level accuracy can be achieved. The core measurement mechanism lies in precisely reconstructing the geometric relationship between the incident laser and reflected optical path, combined with high-speed signal processing techniques to real-time decode the displacement information of the measured target.

In summary, various non-contact measurement modalities, including acoustic, thermal imaging, laser vibrometry, high-speed vision, radar, and electrostatic sensors, each offer distinctive advantages. Through advanced data fusion and signal processing methods, the reliability and interpretability of fault diagnosis can be significantly enhanced. Subsequent sections will systematically elaborate on the typical analysis methods for signals acquired by the aforementioned sensor categories.

3. Non-contact fault diagnosis methods

In recent years, non-contact fault diagnosis has garnered widespread attention from both academia and industry, as it enables real-time acquisition of high-fidelity fault information without disrupting equipment operation. However, raw signals from acoustic, thermal, and optical sensors have low SNR, non-stationarity, and high dimensionality, posing substantial challenges for accurate fault diagnosis. Traditional approaches relying on handcrafted time-frequency features and shallow classifiers often struggle to generalize across varying operating conditions and noise levels, primarily due to their limited capacity to capture hierarchical and nonlinear fault signatures. Recent advances in

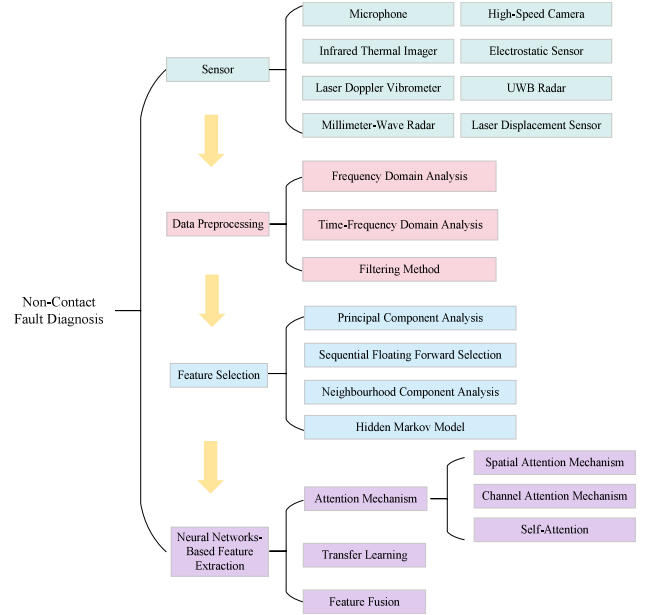


Fig. 2. Overview of the architecture.

neural network methods have markedly pushed the state of the art by enabling end-to-end learning of discriminative representations directly from raw or minimally preprocessed non-contact data. Specifically, deep architectures equipped with attention mechanisms dynamically focus on fault-relevant frequency bands or spatial regions, effectively suppressing strong background interference. Transfer learning allows diagnosis knowledge acquired from one sensor modality or operating condition to be robustly adapted to another, mitigating data scarcity and domain mismatch. Furthermore, feature fusion strategies within deep networks integrate complementary information from multiple non-contact sensors, enhancing fault discriminability and resilience. Overall, these neural network advances overcome the limitations of conventional feature engineering, delivering substantial gains in diagnostic accuracy, cross-domain generalization, and interpretability. To systematically elucidate these advances, this section first reviews signal preprocessing and statistical feature extraction techniques, then delves into neural network architectures including attention mechanisms, transfer learning, and feature fusion, and finally discusses their efficacy and limitations in advancing non-contact fault diagnosis.

3.1. Data preprocessing for deep neural networks

The overall processing pipeline of non-contact fault diagnosis is depicted in Fig. 2. As shown, data preprocessing serves as the first critical stage. Signal processing technology, as a core methodology in modern information processing, draws its theoretical foundations from the interdisciplinary integration of Fourier analysis, stochastic processes, and information theory (Hou et al., 2026). Through systematic algorithms for signal acquisition, analysis, transformation, and reconstruction, it provides crucial feature extraction and data augmentation capabilities for deep learning models. Signal processing techniques are paramount to both the theoretical advancement and practical implementation of non-contact fault diagnosis. Multi-modal data acquired by non-contact sensors (such as array microphones, infrared thermal imagers, and laser Doppler vibrometers) exhibit inherent non-stationarity, low SNRs, and cross-domain coupling characteristics, necessitating rigorous preprocessing workflows to meet the input requirements of deep learning models. This workflow follows a three-stage paradigm comprising noise suppression, feature enhancement, and dimension alignment.

3.1.1. Frequency domain analysis methods

In non-contact fault diagnosis scenarios, frequency-domain analysis has become a core paradigm for processing multi-source heterogeneous information including acoustic, visual, and electromagnetic data, owing to its sensitivity to fault feature frequencies and inherent physical interpretability. This paper systematically reviews the differentiated application strategies of Fourier transform family algorithms, including Discrete Fourier Transform (DFT), Fast Fourier Transform (FFT), and spectral derivative methods, in non-contact diagnosis, and further explores the necessity and feasible pathways for extending these algorithms toward time-frequency joint analysis.

DFT is a mathematical algorithm used to convert discrete-time signals from the time domain to the frequency domain. As a specific application of Fourier transform in the discrete signal domain, DFT is used to precisely analyze the spectral components of the signal, thereby revealing and quantifying the frequency components contained within the signal. For a discrete signal sequence $x[n]$ acquired by non-contact sensors, the DFT is defined as

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j \frac{2\pi}{N} kn} \quad (1)$$

where $X[k]$ represents the complex amplitude of the k -th frequency component of signal $x[n]$, k is the frequency index ranging from 0 to $N-1$, j denotes the imaginary unit, and $e^{-j \frac{2\pi}{N} kn}$ is the complex exponential function representing a complex exponential wave at frequency $\frac{2\pi}{N} k$.

FFT (Cooley & Tukey, 1965) is an efficient algorithmic implementation of the Discrete Fourier Transform, which reduces computational complexity from $O(N^2)$ to $O(N \log N)$ through the Cooley-Tukey algorithm, thereby enabling real-time high-resolution spectral analysis on embedded devices. Compared with contact-based diagnosis, non-contact diagnosis exhibits greater sensitivity to FFT parameter configuration. Taking acoustic diagnosis as an example, when suppressing environmental broadband noise such as aerodynamic noise and electromagnetic interference, sampling rates of 1–10 kHz and 1024–4096-point FFT are typically employed. For laser Doppler vibrometry signals aimed at capturing weak high-frequency impacts, however, sampling rates above 50 kHz and 16384-point FFT are required to achieve 0.6 Hz frequency resolution within the 1–10 kHz band.

Lu et al. (2012) first transformed the sound pressure signal into spectrograms via FFT, and then employed the Gray-Level Co-occurrence Matrix to quantify spatial features such as spectral texture contrast and entropy, thereby capturing and characterizing potential fault features. Based on this, Lu et al. (2013) further employed FFT to obtain spectra and spectrograms of gearbox sound fields under varying operating conditions, extracting spatial distribution features of acoustic field images through texture analysis to achieve fault identification. Seo et al. (2014) converted time-domain signals to the frequency domain via FFT, combining moving detection methods for periodic impact sound features to realize fault diagnosis. Liu et al. (2020) extracted displacement information from vibration images under different fault conditions and employed FFT to obtain their frequency-domain features. Subsequently, a linear model was constructed to establish the mapping relationship between vibration features and fault types. Cristalli et al. (2006) used the FFT to convert the displacement signals obtained by the laser vibration meter into spectrogram forms. Then, they integrated the spectral energy within a specific time window to construct feature vectors highly related to stator eccentricity faults. It is worth noting that for non-contact vibration signals with low-frequency shift, the exponential window is superior to the Hanning window to prevent the base frequency leakage from masking harmonic features. Chen et al. (2024a) applied FFT to convert non-contact vibration signals measured by LDV from the time domain to the frequency domain, thereby enabling more efficient extraction of characteristic frequencies and their amplitudes from bearing vibration signals, consequently facilitating precise fault feature identification. However, conventional FFT is constrained by the Heisenberg

uncertainty principle when analyzing transient bearing impacts, exhibiting an inherent limitation in the time-frequency resolution trade-off. To address this, Park et al. (2021) proposed a frequency energy shifting method, enabling early crack identification in wind turbine blade bearings through tracking the migration of energy centers within fault frequency bands.

Frequency-domain analysis has established a comprehensive technical framework encompassing FFT, spectral feature extraction, and fault mapping, and has been widely adopted for non-contact fault diagnosis. Nevertheless, applicability to non-stationary signals remains limited, as manifested in three primary aspects. First, time-varying frequency content is difficult to characterize. Second, sensitivity to transient impulsive components is insufficient. Third, time-frequency resolution is fundamentally constrained by the uncertainty principle. To address these limitations, time-frequency analysis methods such as the Short-Time Fourier Transform (STFT) and Wavelet Transform have been proposed, enabling precise characterization of non-stationary fault signals through joint time-frequency representation.

3.1.2. Time-frequency domain analysis methods

Time-frequency analysis techniques establish a novel paradigm for multidimensional signal interpretation in non-contact fault diagnosis by constructing joint time-frequency representations. This methodological framework offers three core theoretical advantages: (i) accurate capture and dynamic extraction of time-varying features in nonstationary signals, (ii) effective separation of target signals from complex background noise via time-frequency filtering mechanisms, and (iii) adaptive optimization of the time-frequency resolution trade-off, thereby overcoming limitations of conventional Fourier analysis. Notably, the time-frequency localization capability enables microscale analysis of signal transients, rendering it particularly suitable for detecting incipient weak faults in rotating machinery. Furthermore, the synergistic integration of time-frequency representations with deep neural networks provides a robust framework for fault discrimination in non-contact sensing signals, such as acoustic and infrared measurements. This technical approach has demonstrated promising application prospects in the field.

STFT (Gabor, 1946) serves as a classical linear time-frequency analysis tool that achieves local spectral estimation of non-stationary signals through sliding window functions. In non-contact diagnosis scenarios, its performance is significantly constrained by window length selection. For acoustic signals containing broadband environmental noise, an adaptive window length strategy is required, employing short windows for high-frequency bands and long windows for low-frequency bands, thereby balancing time-frequency resolution within the 1–5 kHz fault-related frequency band while suppressing periodic interference generated by rotating machinery. Conversely, for infrared thermal image sequences where fault-induced temperature variations evolve relatively slowly, fixed long windows may be adopted to enhance low-frequency energy concentration. The STFT is formulated as follows.

$$X(t, f) = \int_{-\infty}^{+\infty} x(\tau) \omega(\tau - t) e^{-j2\pi f \tau} d\tau \quad (2)$$

where $X(t, f)$ represents the STFT result at time t and frequency f , $x(\tau)$ is the original signal, $\omega(\tau - t)$ denotes the window function, typically centered at t , and $e^{-j2\pi f \tau}$ is the complex exponential function for Fourier transformation.

Zhang et al. (2023b) proposed a transfer convolutional recurrent network that employs STFT to map acoustic signals into the time-frequency domain, utilizing pre-trained models to alleviate data imbalance issues. However, the time-frequency resolution trade-off characteristic of STFT constrains its performance, wherein shortening the analysis window enhances temporal resolution at the expense of frequency resolution. Consequently, the efficacy of this approach diminishes significantly for signals containing rapid transient high-frequency components.

Wavelet Transform (Grossmann & Morlet, 1984) demonstrates superior transient feature extraction capability in non-contact diagnosis

through its multi-resolution analysis framework. Specifically, Morlet wavelets with center frequency of 1 kHz and bandwidth parameters of 6–8 are appropriate for acoustic signal processing, as this wavelet class can effectively match the damped oscillatory waveforms excited by bearing fault impacts. Conversely, infrared thermal image analysis favors low-order Daubechies wavelets (e.g., db2) with three-level decomposition to preserve the spatial continuity of temperature fields. Barusu et al. (2019) proposed an adaptive extended wavelet transform method that decomposes ultra-wideband radar signals into eight scales, combined with Bayesian threshold estimation, achieving detection of 0.1 mm cracks in early motor bearing fault diagnosis. The mathematical formulations of wavelet forward and inverse transforms are presented as follows.

$$CWT_f(t) = \frac{1}{\sqrt{|\psi(\tau)|}} \int_{-\infty}^{+\infty} f(t) \frac{\psi^*\left(\frac{t-\tau}{\sigma}\right)}{\sigma} d\tau \quad (3)$$

where $f(t)$ is the original signal, ψ represents the wavelet function, σ is the scale parameter, τ is the translation parameter, and ψ^* denotes the complex conjugate of the wavelet function.

$$\hat{f}(t) = \frac{1}{C} \int_a^b \int_{-\infty}^{+\infty} d(\sigma, \tau) \frac{\psi\left(\frac{\tau-t}{\sigma}\right)}{\sigma} d\sigma d\tau \quad (4)$$

where $d(\sigma, \tau)$ represents the processed wavelet coefficients, C is the normalization constant (admissibility condition), and the integration limits a and b define the bounds of the scale parameter.

Wavelet Transform exhibits remarkable advantages in non-contact fault diagnosis of rotating machinery. These advantages stem from its superior time-frequency localization characteristics. In particular, it excels at extracting weak transient impact features from strong background noise. Such features are indicative of localized faults, including bearing spalling and gear tooth fracture. Nevertheless, this technique suffers from a critical limitation in that diagnosis accuracy is highly sensitive to the correspondence between signal characteristics and specific fault types (e.g., bearing spalling impacts versus gear wear modulation sidebands). Divergent characteristic patterns across fault types often result in substantial performance fluctuations. More importantly, the selection of wavelet basis functions (such as the compact support of Daubechies wavelets and the frequency concentration of Morlet wavelets) exerts decisive influence on both impact feature extraction precision and noise suppression effectiveness, necessitating directed optimization with respect to target fault impact features (pulse width and frequency band range) and background noise properties.

Wavelet denoising (Donoho, 1995; Donoho & Johnstone, 1994; Mallat, 1989), as an advanced signal enhancement technique based on wavelet transform, serves a critical preprocessing function in non-contact diagnosis. This technique maps disturbed complex signals acquired by non-contact sensors into wavelet coefficient spaces across different scales and frequency bands through multi-scale decomposition, thereby achieving preliminary separation by exploiting significant differences in energy distribution and time-frequency concentration between fault features and background noise. Furthermore, adaptive threshold methods tailored to non-contact signal characteristics (such as low SNR and non-stationary noise) are employed to precisely suppress noise-dominated coefficients while maximally preserving key coefficients containing weak fault information. The aforementioned processing provides high-quality, high SNR input signals for subsequent refined fault feature extraction (e.g., envelope spectrum analysis) and intelligent diagnostic algorithms, constituting a crucial link in enhancing the reliability of non-contact diagnosis. The relevant formulas for wavelet denoising can be expressed as

$$d_{j,k} = \int_{-\infty}^{+\infty} f(t) \psi_{j,k}(t) dt \quad (5)$$

$$\hat{d}_{j,k} = \begin{cases} d_{j,k}, & |d_{j,k}| \geq \lambda \\ 0, & |d_{j,k}| < \lambda \end{cases} \quad (\text{Hard}) \quad (6)$$

$$\hat{d}_{j,k} = \begin{cases} \text{sgn}(d_{j,k})(|d_{j,k}| - \lambda), & |d_{j,k}| \geq \lambda \\ 0, & |d_{j,k}| < \lambda \end{cases} \quad (\text{Soft}) \quad (7)$$

$$\lambda = \sigma \sqrt{2 \ln N} \quad (8)$$

$$\hat{f}(t) = \sum_{j,k} \hat{d}_{j,k} \psi_{j,k}(t) \quad (9)$$

where $f(t)$ denotes the original signal, $\psi_{j,k}(t)$ represents the wavelet basis functions, $d_{j,k}$ are the coefficients obtained from wavelet decomposition, $\hat{d}_{j,k}$ are the thresholded wavelet coefficients, λ is the threshold, typically set to $\sigma \sqrt{2 \ln N}$ (universal threshold), $\hat{f}(t)$ is the denoised signal, σ indicates the noise standard deviation, and N is the signal length.

Mishra et al. (2017) demonstrated that, for low-speed rotating machinery, the synergy between wavelet denoising and Sigmoid-function-based threshold envelope analysis enables effective extraction of fault-related features from non-contact vibration signals acquired by laser vibrometers, thereby establishing a robust signal processing framework for bearing fault diagnosis. Building upon this paradigm, Jung et al. (2021b) further extended this approach to acoustic sensing scenarios, where the integration of adaptive wavelet denoising, statistical spectral acoustic features, and one-dimensional convolutional neural networks significantly enhanced the precision of fault pattern recognition, diagnosis accuracy, and system automation.

In non-contact fault diagnosis, environmental noise and complex transmission paths often render observed signals non-stationary and low SNR. The Discrete Wavelet Transform (DWT) discretely samples the mother wavelet through dyadic scale-translation grids. It preserves the time-frequency localization advantages of the continuous wavelet transform while significantly reducing computational complexity via the Mallat algorithm, and ensures energy conservation through orthogonal or biorthogonal basis functions. Consequently, DWT has become an effective tool for processing non-contact observation data such as rotating machinery vibration and infrared thermography. In the field of infrared thermography, Younus and Yang (2012) first employed two-dimensional DWT for multi-resolution decomposition, combining Mahalanobis distance with the Relief algorithm to construct low-dimensional feature vectors, significantly improving fault classification accuracy. Choudhary et al. (2018) similarly utilized two-dimensional DWT to preprocess thermographic images, selecting optimal features based on Mahalanobis distance and feeding them into an SVM classifier for bearing fault identification. In vibration signal processing, Goyal et al. (2020a) applied DWT for denoising non-contact vibration signals acquired by their developed low-cost laser-based sensor, employed Mahalanobis distance to select the most discriminative features, and fed them into an SVM to complete diagnostic decisions.

To compress feature dimensions and suppress redundant information, Choudhary et al. (2021a) introduced Principal Component Analysis (PCA) following two-dimensional DWT decomposition. In this method, high-dimensional wavelet coefficients were first subjected to linear dimensionality reduction, after which Mahalanobis distance was employed to evaluate the correlation between principal components and fault categories, thereby achieving secondary feature selection. To address stripe noise and background interference in infrared thermography, Kala et al. (2024) proposed a hybrid framework combining two-dimensional DWT with Independent Component Analysis (ICA). First, ten normalized statistical features were extracted from bearing thermal images. Subsequently, two-dimensional DWT was utilized to decompose the features into different frequency and resolution levels. Then, independent component features were extracted through ICA. Finally, feature importance ranking was performed based on Mahalanobis distance to construct a critical feature matrix for bearing fault diagnosis.

Wavelet Packet Decomposition (WPD) (Torrésani, 1992) is an extension of wavelet theory that achieves complete characterization of time-frequency energy through an adaptive full-band decomposition mechanism. Unlike traditional wavelets that recursively partition only low-frequency components, WPD can continue to subdivide high-

frequency subbands at any node, exhibiting stronger capture capability for transient features in non-contact acoustic signals. Kang et al. (2016) proposed a compound fault diagnosis strategy based on transient acoustic signals and rough set theory. They employed db6 wavelet for WPD, extracted four feature vectors after normalizing subband energies, and subsequently performed attribute reduction through rough set theory, effectively improving the accuracy and efficiency of bearing fault diagnosis. Guang et al. (2023) integrated non-contact vibration data captured by an event camera into cumulative event frames, extracted time-domain signals through Gabor filtering, and then conducted three-level WPD and envelope spectrum analysis to obtain fault feature frequencies. Notably, WPD is capable of further subdividing high-frequency detail subbands into fault feature frequency bands (e.g., bearing outer race fault feature frequency and its harmonics). Its entropy indicators exhibit sensitivity to modulation phenomena in acoustic signals, thereby providing effective features for compound fault diagnosis of wind turbine gearboxes. Nevertheless, the selection of wavelet basis and decomposition levels still depends on prior knowledge, restricting its generalization performance under complex operating conditions.

Empirical Mode Decomposition (EMD) (Huang et al., 1998) provides a data-driven adaptive time-frequency decomposition paradigm. EMD decomposes complex waveforms into several Intrinsic Mode Functions (IMFs) based on the local extremum scales of the signal, requiring no pre-specified basis functions and exhibiting adaptability to nonlinear and non-stationary processes. In contact-based vibration analysis, EMD is commonly employed to separate transient impact patterns. Specifically, the first three high-frequency IMFs within the 1–10 kHz range are extracted to effectively suppress motor fundamental frequencies and broadband noise. Clear time-frequency ridge lines are then generated through Hilbert transform, highlighting intermittent impact features caused by cracks or spalling. However, environmental interference in non-contact signals is more complex, requiring adjusted EMD application strategies. For infrared temperature signals, mid-order IMFs within the 0.1–1 Hz range are retained to reflect temperature gradient variations, while high-frequency measurement noise and low-frequency ambient temperature drift are filtered out. For acoustic signals, decomposition into 8–12 IMFs is performed, and effective modes containing 1–5 kHz fault acoustic features are selected through energy ratio screening. High-order IMFs associated with airflow and equipment interference are suppressed by this method, and the acoustic signals are ultimately reconstructed in a purified form focused on bearing regions.

EMD is widely applied in non-contact fault diagnosis. To suppress mode mixing and noise interference, Vanraj et al. (2017) introduced complete ensemble EMD with adaptive noise into the acoustic diagnosis process. To address the problem of inaccurate rolling bearing fault diagnosis caused by low SNR in acoustic data, Xu et al. (2020) proposed a three-stage method. First, EMD was employed to decompose the raw signal into multiple IMFs. Second, fault-sensitive IMFs were selected using a sensitivity evaluation algorithm, and then refined into finite-bandwidth local IMFs through Variational Mode Decomposition (VMD). Finally, local IMFs were reorganized based on signal characteristics to accurately extract fault features, effectively improving weak fault diagnosis accuracy. Yin et al. (2022) proposed an EMD-based multi-feature fusion method for acoustic signals to improve diagnosis accuracy of nonlinear and non-stationary signals. First, EMD was adopted to obtain multi-scale decomposition results. Second, the IMF that best reflected the fault state was selected according to the maximum kurtosis criterion. Finally, fusion features with minimal correlation were extracted through Pearson correlation coefficient analysis.

EMD-based non-contact fault diagnosis methods performs well in mechanical condition monitoring. However, when dealing with strong non-stationary signals, mode mixing and end effects significantly reduce diagnosis accuracy. To overcome these limitations, existing research employs VMD to achieve precise separation of signal frequency com-

ponents. VMD was introduced by Dragomiretskiy and Zosso in 2014 (Dragomiretskiy & Zosso, 2013) with the aim of mitigating mode mixing in EMD and enhancing computational efficiency and applicability for multi-component signal processing. Grounded in the variational principle, this approach decomposes signals by minimizing inter-modal correlations and promoting modal locality. However, parameter configurations and implementation strategies differ markedly between contact-based and non-contact diagnostic approaches. Contact diagnosis focuses on precise separation of high-frequency transient impact modes, typically setting 5–8 modes, larger penalty factors, and low noise tolerance, screening effective modes by matching fault feature frequencies. Non-contact diagnosis emphasizes anti-interference extraction of weak feature modes, adopting 3–6 modes, smaller penalty factors, and higher noise tolerance, requiring combination of energy ratio and correlation analysis to distinguish fault-related signals from environmental interference. The schematic flow of VMD is shown in Fig. 3, and its core variational optimization problem is formulated as follows.

$$\min_{\{u_k\}, \{\omega_k\}} \left\{ \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \quad (10)$$

$$\text{s.t.} \quad \sum_{k=1}^K u_k(t) = f(t)$$

where $\{u_k\}$ represents the K IMFs obtained from the decomposition, $\{\omega_k\}$ denotes the center frequencies of respective IMFs, $\delta(t)$ is the Dirac delta function, $\frac{j}{\pi t}$ represents the Hilbert transform kernel, together they constitute the bandwidth constraint operator, $f(t)$ is the original signal to be decomposed. The optimization objective entails minimization of the total bandwidth energy associated with each IMF, subject to the exact reconstruction constraint requiring that all IMFs sum to the original signal.

To further enhance the accuracy and engineering applicability of non-contact fault diagnosis under complex operating conditions, researchers have conducted sustained and in-depth theoretical improvements and algorithmic fusion centered on the VMD framework. Shan et al. (2023) proposed a motor bearing fault acoustic pattern recognition method based on the Mel-CNN model. First, VMD technology was employed to eliminate high-frequency noise interference and extract Mel-spectral features. Subsequently, convolutional neural networks were utilized to further extract high-dimensional abstract features, achieving accurate bearing fault identification. To address the issue of weak early fault features in rotating bearings of chemical pumps, Cao et al. (2023) proposed an improved VMD fault feature extraction method. They employed the Sparrow Search Algorithm to optimize VMD parameters and combined it with Hilbert transform to screen modal components rich in fault information, significantly enhancing decomposition performance and feature extraction effectiveness. Despite exhibiting good adaptability in processing complex non-stationary signals, VMD-based non-contact fault diagnosis methods face substantial computational resource challenges due to their high computational complexity when dealing with large-scale datasets.

To address the challenge of extracting impact periodic features in early fault diagnosis of rotating machinery, the Feature Mode Decomposition (FMD) (Miao et al., 2023) method is proposed. By organically integrating the impact periodic feature extraction mechanism with the signal decomposition framework, an adaptive finite impulse response filter bank architecture is established. This architecture dynamically adjusts the filter coefficients based on the impact period coupling characteristics present in non-contact monitoring signals of rotating machinery, thereby achieving high-precision separation and directional feature extraction of fault modal components under strong background noise and composite fault conditions, significantly improving the robustness of non-contact diagnosis. Zhang et al. (2023a) introduced a fault bearing acoustic localization method grounded in cyclostationary beamforming.

Table 2
Advantages and limitations of time-frequency analysis methods.

Method	Advantages	Limitations
STFT	Simple, physically interpretable	Fixed time-frequency resolution trade-off
Wavelet Transform	Multi-resolution, excellent for weak transients	Basis function selection critical, fault type sensitive
Wavelet Denoising	Effective noise suppression	Threshold selection problem
DWT	Low complexity	Low frequency resolution at high bands
WPD	Full-band decomposition, captures high-frequency details	Decomposition level and basis rely on prior knowledge
EMD	Fully data-adaptive, no basis needed	Mode mixing, end effects
VMD	Mode mixing overcome, controllable bandwidth	High computational cost, parameter tuning
FMD	Robust to heavy noise, periodic impact extraction	Parameter standardization needed
RSSD	Separation of periodic and sparse transient components	Regularization parameters empirical

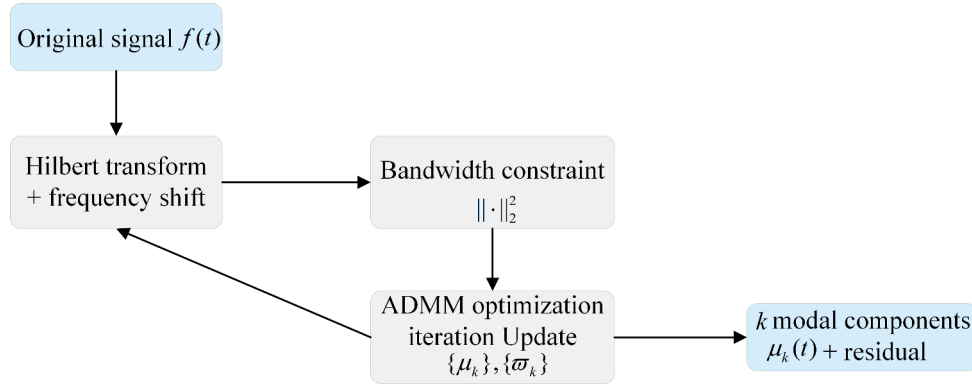


Fig. 3. Schematic diagram of the VMD process.

Frequency information obtained via FMD was employed to prioritize the analysis of cyclic spectral density over the scanned plane of interest, enabling precise localization of the fault-bearing component.

Resonance-based Sparse Signal Decomposition (RSSD) is an advanced signal processing method for non-contact fault diagnosis of rotating machinery. Based on a joint optimization framework, this method adaptively decomposes complex vibration and acoustic signals into high-resonance components and sparse impulse components. High-resonance components characterize the periodic vibration features of rotating machinery systems (e.g., gear meshing frequencies, shaft rotational frequencies, and their harmonics). Sparse transient components, in contrast, accurately extract weak impulsive responses induced by localized faults, providing an effective feature separation means for non-contact diagnosis and proving critical for detecting and identifying early weak faults under strong noise backgrounds. Qiu et al. (2023) employed sparse representation methods to filter background noise from Doppler-corrected acoustic signals and reconstructed noise-free signals through inverse wavelet transform. Subsequently, RSSD technique was applied to remove interfering high-order harmonic signals from the signal. Finally, genetic algorithm was utilized to optimize the parameters of the radial basis function neural network. The RSSD formulation is expressed as follows:

$$x(t) = \sum_k r_k(t) + \sum_k s_k(t) \quad (11)$$

where $x(t)$ denotes the original signal, $r_k(t)$ represents the k -th resonance component, and $s_k(t)$ represents the k -th sparse component. The optimization objective seeks to minimize the energy of sparse components while retaining the high-resonance properties inherent to periodic components, which is mathematically formulated as:

$$\min_{r_k, s_k} \|x(t) - \sum_k r_k(t) - \sum_k s_k(t)\|_2^2 + \lambda_1 \sum_k \|r_k(t)\|_{H_1} + \lambda_2 \sum_k \|s_k(t)\|_1 \quad (12)$$

where $\|\cdot\|_{H_1}$ denotes the high-resonance norm (measuring periodicity through ℓ_1 -norm of autocorrelation), $\|\cdot\|_1$ denotes the sparsity-promoting ℓ_1 -norm, and λ_1, λ_2 are regularization parameters balancing

the trade-off between reconstruction fidelity and component characteristics. The key advantages and limitations of the aforementioned time-frequency analysis methods are summarized and compared in Table 2.

3.1.3. Filtering methods

Signal filtering serves as a critical technique in non-contact fault diagnosis for suppressing environmental noise and highlighting weak fault features. Among various approaches, spatial domain filtering based on array signal processing theory leverages coherent accumulation of multi-channel observation data across spatial dimensions, providing spatial selectivity gain for the diagnosis system. As a representative implementation of spatial domain filtering, beamforming optimizes the phase-amplitude response of sensor arrays, forming high-gain main lobes in target directions and deep nulls in non-target directions to suppress interference and noise. This significantly improves the SNR under strong background noise conditions, overcoming the performance limitations of traditional single-point measurements. Shuai et al. (2023) introduced the minimum variance distortionless response (MVDR) beamformer into the bearing acoustic emission detection. They combined it with the multi-point optimal minimum entropy deconvolution to extract pulse signals from the beamformer output, thereby achieving the enhancement and identification of weak fault features. Furthermore, Wang et al. (2024) proposed a non-contact multi-point bearing fault diagnosis method based on multi-input multi-output frequency-modulated continuous-wave radar. In this method, an improved MVDR beamforming method was adopted to reconstruct the covariance matrix of interference and noise, thereby significantly improving the beam pattern performance. To address the issue of severe degradation in diagnosis accuracy caused by strong noise features, Yuan et al. (2024) proposed a spatial filtering algorithm based on MVDR and Generalized Sidelobe Canceller, achieving directional noise suppression. The relevant beamforming formulations are expressed as follows:

$$X(\omega) = w^H S(\omega) \quad (13)$$

where $X(\omega)$ denotes the frequency-domain representation of the filtered signal, w^H represents the Hermitian transpose of the weight vector, and $S(\omega)$ denotes the frequency-domain representation of signals received by the sensor array.

The optimal weight vector for MVDR beamforming is obtained through constrained optimization:

$$\min_w w^H \mathbf{R}_{nn} w \quad \text{subject to} \quad w^H \mathbf{a}(\theta_0) = 1 \quad (14)$$

where $\mathbf{R}_{nn}$ represents the noise-plus-interference covariance matrix, $\mathbf{a}(\theta_0)$ denotes the steering vector toward the target direction θ_0 , and the constraint ensures distortionless response in the desired look direction while minimizing output variance.

Maximum Correlated Kurtosis Deconvolution (MCKD) (McDonald et al., 2012) is a higher-order statistics-based signal enhancement method that optimizes a Finite Impulse Response (FIR) filter to maximize the correlated kurtosis of the output sequence. This metric exhibits inherent sensitivity to periodic transient impulses, making it particularly suitable for extracting non-stationary features caused by incipient faults in rotating machinery. In non-contact diagnosis scenarios, MCKD effectively enhances the SNR of acoustic and vibration signals by amplifying fault-related periodic impact components. Zhang et al. (2022b) proposed an improved MCKD technique by calculating the autocorrelation of the acoustic signal to estimate the iteration period, thereby reducing the number of required input parameters and eliminating the need for resampling. The improved MCKD algorithm is formulated as

$$\kappa = \frac{E[(y - \mu)^4]}{\sigma^4} \quad (15)$$

$$\mathbf{R}_{xy}(\tau) = \int_{-\infty}^{+\infty} x(t)y(t + \tau)dt \quad (16)$$

where κ denotes the kurtosis, μ represents the mean of the signal, σ represents the standard deviation of the signal, $R_{xy}(\tau)$ is the cross-correlation function, with $x(t)$ and $y(t)$ denoting the signals.

After noise suppression and signal enhancement through preprocessing techniques such as time-frequency analysis and spatial filtering, the obtained data may still be high-dimensional and contain redundancy. Thus, in the next section, we will discuss statistical techniques for reducing dimensionality and selecting discriminative features to achieve efficient and accurate fault diagnosis.

3.2. Feature selection for deep learning

Statistical techniques serve as a critical bridge between raw sensor data and deep neural networks, with their core role lying in transforming high-dimensional, noisy non-contact monitoring signals into compact and highly discriminative feature representations. In the preprocessing pipeline of neural network-based fault diagnosis, statistical methods primarily undertake two essential functions. One is dimensionality reduction, which projects multi-source high-dimensional redundant data from acoustics, vibration, and thermography into low-dimensional subspaces while preserving discriminative information closely related to fault diagnosis. The other is feature selection, which identifies and extracts optimal feature subsets capable of maximizing inter-class separability. By filtering out noise-induced spurious correlations and extracting critical diagnostic information from massive monitoring data, statistical preprocessing effectively guides neural networks to focus on robust fault-relevant patterns rather than dataset-specific interference artifacts, thereby significantly enhancing the model generalization performance across varying operating conditions and sensor configurations.

To achieve effective dimensionality reduction while preserving diagnosis-relevant information, PCA has emerged as a cornerstone technique for compressing high-dimensional non-contact monitoring data (Hotelling, 1933; Pearson, 1901). PCA projects high-dimensional correlated variables onto a low-dimensional subspace spanned by principal components through orthogonal linear transformation, preserving

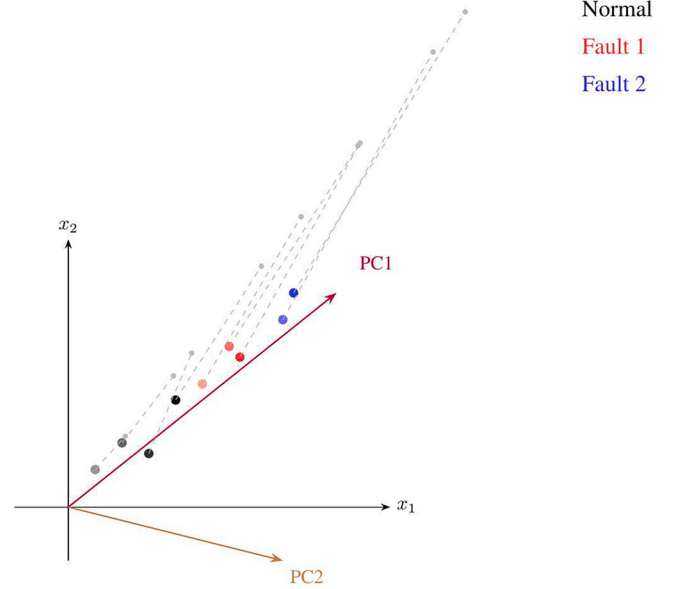


Fig. 4. Schematic diagram of fault diagnosis based on 2D PCA.

the dominant energy of original data under the variance maximization criterion. For high-dimensional data obtained from non-contact monitoring, including multi-channel vibration, acoustic, or thermal imagery, redundant information often dominates. PCA concentrates energy into the first few principal components via eigen-decomposition of the covariance matrix, thereby compressing data dimensionality by one to two orders of magnitude with minimal loss of diagnosis information, significantly reducing computational complexity for subsequent classifiers or regression models. In Choudhary et al. (2021a), Goyal et al. (2020b, 2021), PCA was used for feature reduction in non-contact fault diagnosis, effectively improving diagnosis accuracy, as illustrated in Fig. 4.

Although PCA achieves efficient dimensionality reduction, its unsupervised nature may suppress features that are highly correlated with fault categories yet exhibit low variance. To mitigate this limitation, supervised feature selection methods such as Sequential Forward Floating Selection (SFFS) have been widely adopted. SFFS is a statistical method that iteratively adds and removes features to identify the minimal optimal subset with maximal discriminability from high-dimensional spaces. In non-contact fault diagnosis, raw data often contains numerous features, but not all contribute equally to diagnostic performance. In vibration-based fault diagnosis, SFFS can precisely identify frequency features strongly associated with specific fault modes while eliminating irrelevant or redundant ones, thereby significantly improving model accuracy across different fault types. In Goyal et al. (2020b, 2021), SFFS was used to select the most representative and discriminative feature subset, which improved the accuracy of non-contact fault diagnosis.

The complex local manifold structures exhibited by different fault categories in the original high-dimensional space render linear dimensionality reduction methods ineffective in preserving local class separability. A Mahalanobis distance metric is learned via Neighborhood Components Analysis (Goldberger et al., 2004) by maximizing the expected leave-one-out classification accuracy in a stochastic nearest-neighbor setting, thereby ensuring that samples from the same class remain compact in the low-dimensional embedding space while those from different classes are well separated. This metric learning method is particularly effective for non-contact modalities (such as thermal imaging), where high spatial resolution and noise levels coexist, and the temperature difference between the faulty area and the normal area depends on the local neighborhood relationship for correct distinction. For instance, a thermal imaging-based non-contact diagnosis method was proposed by

Choudhary et al. (2021b), in which dimensionality reduction was performed via NCA with optimal feature weighting computed for each feature, effectively enhancing fault diagnosis accuracy. Beyond distance metric learning, an alternative paradigm is offered by probabilistic state-space models for capturing temporal dynamics in diagnostic signals. A statistical framework is constituted by Discrete Hidden Markov Models (Baum et al., 1970) for describing systems capable of transitioning among multiple hidden states, each associated with a probability distribution over observable emissions.

3.3. Neural networks

Data preprocessing and statistical feature extraction techniques form the foundational layer of neural network-based fault diagnosis. Through noise reduction, dimensionality reduction, and structured representation, raw high-dimensional monitoring signals are transformed into a low-dimensional discriminative feature space, thereby enabling deep models to more effectively focus on fault-related patterns rather than the noise in the data. Nevertheless, the efficacy of such handcrafted feature engineering is fundamentally bounded by its limited capacity to capture hierarchical and nonlinear fault features concealed in complex sensor data, especially under variable operating conditions and strong environmental interference. To transcend these limitations, deep neural networks have emerged as the dominant paradigm, offering end-to-end learning capabilities that automatically discover task-relevant hierarchical features from preprocessed inputs with superior representational power. This section systematically reviews neural network architectures tailored for non-contact fault diagnosis, encompassing deep feature learning frameworks, attention mechanisms for adaptive feature weighting, transfer learning strategies for cross-domain knowledge reuse, and feature fusion strategies that integrate multi-source heterogeneous information for enhanced diagnosis performance.

3.3.1. Attention mechanisms

In non-contact bearing fault diagnosis, acoustic and vibration signals encounter dual challenges during the acquisition process, i.e., high background noise and multi-source mechanical coupling interference, which prevent traditional feature extraction methods from effectively capturing weak fault features under complex operating conditions. To address this issue, the attention mechanism automatically focuses attention on the feature frequencies of vibration and acoustic signals that are directly related to bearing faults, while suppressing irrelevant noise. Thus, it provides robust, time-varying, and interpretable feature representations for variable speeds, variable loads, and different fault types. Through this data-based self-optimization paradigm, the generalization performance of diagnostic models in industrial noise environments has been significantly improved.

In recent years, considerable momentum has been gained by fault diagnosis approaches based on the attention mechanism, with their significance becoming increasingly prominent in handling complex and high-dimensional datasets. On one hand, the contribution of individual features can be explicitly quantified by attention weights, enabling diagnostic models to focus on transient impacts in vibration signals or localized temperature rise regions in thermal imaging, thereby providing interpretable evidence for fault traceability. On the other hand, deep fusion of multi-source non-contact data is achieved by cross-modal attention mechanisms through joint modeling of acoustic radiation fields and thermodynamic parameter variations, leading to further improvements in diagnostic accuracy and efficiency. Unlike contact-based fault diagnosis, which relies on high-frequency impact extraction (1–10 kHz), environmental noise pollution in acoustic and thermal imaging spaces must be primarily addressed by non-contact methods. In this situation, the attention mechanism enables mapping between sound source locations or anomalous temperature regions and equipment status, thereby prioritizing the enhancement of critical signal regions while substan-

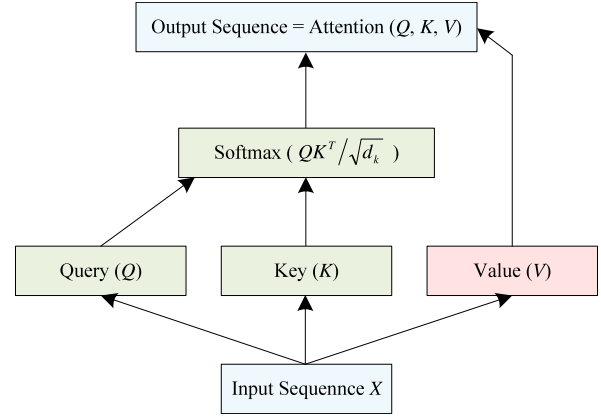


Fig. 5. Schematic diagram of the self-attention mechanism.

tially suppressing background interference. Thus, it is possible to reliably detect weak acoustic responses and thermodynamic deviations.

Non-contact diagnosis with global feature correlation modeling capabilities is provided by the introduction of the self-attention mechanism (Vaswani et al., 2017). Its core lies in the exploitation of Query-Key interactions within the same sequence to capture long-term dependencies, thereby overcoming the limitations of traditional local perception. The schematic diagram of the self-attention mechanism is illustrated in Fig. 5. The discriminative capacity for complex fault patterns is further enhanced by typical extensions such as multi-head self-attention through parallel learning of multiple subspace representations. Xu et al. (2024b) proposed an advanced deep learning model for non-contact bearing fault diagnosis, in which the integration of residual networks and multi-head self-attention mechanisms significantly enhanced capability for rolling bearing fault diagnosis. Yao et al. (2024) proposed an innovative anti-noise method featuring two key components. First, a recursive multi-head self-attention mechanism was designed to adaptively track and suppress actual industrial background noise from multi-channel signals. Second, a spatio-temporal multi-head self-attention mechanism was designed to automatically track remaining noise components after initial suppression and precisely estimate noise interference probability within each signal sample's time-frequency unit.

Channel attention mechanisms achieve intelligent enhancement of fault-sensitive features through establishing dynamic weight allocation models among feature channels. The basic workflow is divided into three steps. First, channel-level statistical descriptors are generated via global average pooling. Second, channel weight vectors are constructed through fully connected layers and nonlinear activation functions (e.g., Sigmoid). Finally, recalibration of feature channels is realized. The main steps encompass global pooling, fully connected layers, activation functions, and reweighting, with the squeezing and excitation network (Hu et al., 2018) being the most widely used. In non-contact fault diagnosis scenarios, due to environmental noise interference and variations in sensor conditions, such as a sensor distance fluctuation of ± 20 cm, the acquired acoustic and vibration signals often exhibit significant uncertainty. Channel attention can dynamically adjust feature weights based on actual data conditions. The spatial attention mechanism mainly works by constructing weight maps that share the same spatial dimension as the input data, and using learning algorithms to encode the spatial importance of input features.

Luo et al. (2023) proposed a rolling bearing fault diagnosis method based on acoustic signals. First, a multi-scale enhanced Top-Hat operator was employed to process input signals through structural elements of different scales, extracting positive and negative pulse features and generating multi-branch data. Second, the generated multi-branch data were weighted through channel attention and spatial attention. Finally, the fused information was fed into neural networks for decision

output. Xu et al. (2024a) integrated channel attention into the InceptionV3 model to enhance feature representation and improve model performance by selectively focusing on information-rich regions in thermal images. Yao et al. (2021) proposed a multi-stage attention mechanism that adaptively estimates time-frequency unit masks by separately learning adjacent information in channel, temporal, and frequency dimensions, multiplying these masks with original spectra to reduce non-smooth noise interference, thereby improving non-contact fault diagnosis accuracy. Peng et al. (2023) designed a texture attention module and an illumination attention module specifically for non-contact vibration data acquired by frame-based cameras. In the texture attention module, spatial information of feature maps was aggregated through average pooling and max pooling operations to extract key texture features. The illumination attention module operated similarly to the texture attention module along the channel dimension but additionally employed a 7×7 convolutional layer to reduce feature mappings to a single channel for capturing illumination information. Finally, these two modules were merged in parallel across two independent dimensions to adaptively optimize intermediate features. Liu et al. (2025a) designed a residual attention pyramid module for mechanical operation sound data, which adaptively learns different levels of fault information through pyramid structure and attention mechanism to enhance feature representation and improve network diagnosis capabilities. Liu et al. (2025b) proposed a multi-channel feature fusion self attention mechanism that integrates the MFCC, spectral, and Mel spectral features of bearing operation sound data, achieving high-precision and strong robustness for non-contact fault diagnosis. Cao et al. (2025) embedded attention mechanisms into graph convolutional and temporal modules, achieving complex spatio-temporal feature interactions and improving non-contact fault diagnosis accuracy.

3.3.2. Transfer learning

In non-contact fault diagnosis tasks, data scarcity and operational complexity impose generalization bottlenecks on conventional supervised learning approaches. Transfer learning offers a systematic solution to these challenges through cross-domain knowledge transfer mechanisms. This section systematically elucidates the paradigmatic value of transfer learning in non-contact fault diagnosis from three dimensions: theoretical essence, technical pathways, and frontier advances. The principal findings encompass three aspects. Firstly, the source to target feature mapping mechanism effectively alleviates the relationship between the distribution changes of fault features and semantic consistency in non-contact sensing data. Secondly, cross distribution transfer technology based on adversarial learning enables diagnostic knowledge to be reused in multi-source non-contact data, overcoming the difficulty of aligning between different distribution data.

The efficacy of transfer learning in non-contact fault diagnosis stems from its capacity to characterize cross-domain invariance of fault features. When acquiring data via non-contact modalities such as vibration accelerometers, acoustic emission sensors, or infrared thermography, raw signals from disparate bearing systems or under varying operational conditions typically exhibit pronounced discrepancies in aspects such as spectral energy distribution and noise levels, which constitute domain shifts. However, the feature frequency components induced by typical faults and their modulation patterns, encompassing rolling element defects and inner race spalling that constitute fundamental fault features, demonstrate consistency across domains. This dichotomy of apparent heterogeneity and intrinsic homogeneity enables cross-domain diagnostic knowledge transfer through the establishment of feature mapping functions from source domains with abundant labeled data to target domains with limited annotations. This process can be formalized as minimizing the discrepancy in feature distributions between domains. Zhiyi et al. (2020) proposed an enhanced convolutional neural network (ECNN) incorporating exponential linear units and stochastic pooling as core architectural components. Convo-

lutional autoencoders were first pre-trained on large-scale unlabeled thermal imaging datasets, with the resulting model parameters subsequently transferred to initialize the ECNN. Fine-tuning was then performed using a small set of labeled thermal images to further optimize the network parameters. Choudhary et al. (2023) first trained a convolutional neural network on source domain data, and then transferred the learned parameters to the target domain for fine-tuning with limited labeled data. In addition, the CORAL loss function was used to minimize the difference in feature distribution between the source domain and the target domain, thereby promoting cross domain fault diagnosis.

Studies (Zhiyi et al., 2020) and Choudhary et al. (2023) address homogeneous modality transfer, while Zhang et al. (2023b) and Li et al. (2024d) extend this to cross-modal scenarios, enhancing model transferability across diverse data types. Specifically, Zhang et al. (2023b) developed a convolutional recurrent neural network within a transfer learning framework, transferring diagnostic knowledge from vibration data of faulty bearings to acoustic-data classifiers, thereby markedly improving diagnosis performance under data imbalance. Furthermore, Li et al. (2024d) proposed a cross-modal feature alignment approach leveraging deep adversarial neural networks to extract modality-invariant features from acceleration and visual data, enabling effective cross-modal knowledge transfer for fault diagnosis.

3.3.3. Feature fusion

A single feature is often insufficient to comprehensively and accurately characterize fault states. Multi-source data, including vibration signals, temperature measurements, acoustic signals, and acoustic emission signals, each reveal critical information regarding bearing operational states. Vibration signals reflect mechanical vibration features, temperature data indicate thermal state variations, and acoustic emission signals capture transient elastic waves generated by incipient micro-damage. Through feature fusion techniques, multi-source heterogeneous information can be integrated to construct comprehensive bearing condition assessment models. This approach fully exploits the complementary advantages of each modality, enabling precise characterization of bearing health status and significantly enhancing the accuracy and reliability of fault diagnosis. This process not only strengthens the robustness of diagnostic systems but also provides multidimensional data support for early fault warning under complex operating conditions. For instance, fusing frequency-domain and time-domain features of vibration signals leverages both fault feature frequencies in the frequency domain and time-domain features such as amplitude and waveform. The discriminative capability of feature vectors among different fault types and severity levels is enhanced, thereby enabling more precise identification of bearing fault modes.

Feature fusion, as a core methodology in machine learning and data mining, primarily addresses the integration of multi-source heterogeneous features. This technique constructs composite feature spaces with higher information density by integrating feature representations derived from different sensors or feature extraction methods, thereby significantly enhancing model representation capacity and predictive performance. Common feature fusion strategies include early fusion, late fusion, and attention-based fusion. Early fusion refers to the integration performed subsequent to feature extraction, with typical operations encompassing concatenation and summation. The concatenation operation connects two or more tensors along a specific dimension, increasing the number of channels or feature dimensions in deep learning models to better capture relationships among diverse features in subsequent layers. The summation operation performs element-wise addition of two tensors to achieve skip connections or residual connections, facilitating gradient flow, improving training stability, and mitigating the vanishing gradient problem in deep networks. The relevant formulas for

concatenation and addition operations are expressed as:

$$Z_{\text{concat}} = \sum_{i=1}^c X_i * K_i + \sum_{i=1}^c Y_i * K_{i+c} \quad (17)$$

$$Z_{\text{add}} = \sum_{i=1}^c X_i * K_i + \sum_{i=1}^c Y_i * K_i \quad (18)$$

where X_i and Y_i represent input channels, $*$ denotes the convolution operation.

Late fusion refers to integrating prediction results from individual models to obtain final predictions. Attention mechanism-based feature fusion methods (Liu et al., 2025b) dynamically adjust feature weights by computing correlations between features, thereby achieving more efficient feature fusion. The integration of features extracted through different signal processing methods significantly improves model diagnosis performance. This fusion approach effectively combines the advantages of individual methods, enabling more comprehensive and in-depth characterization of fault features and consequently enhancing model performance in complex fault diagnosis tasks.

Yin et al. (2022) proposed a multi-step feature extraction framework for acoustic signals. Firstly, the input signal was processed using EMD to obtain multi-scale decomposition results. Then, based on the principle of maximum kurtosis, several IMFs were selected that best represent the fault state. Finally, the fault features were extracted from the selected IMF and the fusion features with the minimum correlation are generated through Pearson correlation coefficient analysis. To address the limitations of single-frame MFCC features and the traditional support vector data description method in noisy environments, Zhang et al. (2023c) first adopted multi-frame fused MFCC features to construct an optimized feature set. Subsequently, robust fault diagnosis was achieved through a sample center distance-weighted support vector data description model. Tang et al. (2023) first extracted seven features for correlation analysis to identify strongly correlated features. Kernel principal component analysis (KPCA) was then applied to fuse these strongly correlated features, and high-precision diagnosis results were ultimately obtained under conditions of high noise and limited samples. An et al. (2023) proposed a non-contact fault diagnosis model based on an improved minimum spatial pyramid pooling. In this model, the conventional sliding window pooling layer was replaced with minimum spatial pyramid pooling, enabling deep fusion of multi-scale fault features while allowing input images of varying scales to be effectively handled. Yao et al. (2024) proposed a dual architecture that combines recursive multi head self-attention mechanism with spatiotemporal multi-head self-attention mechanism. Through the synergistic effect of these two mechanisms, effective fusion of multi-channel acoustic signals was achieved, significantly improving the accuracy and robustness of non-contact fault diagnosis in noisy environments.

Complex industrial environments necessitate the integration of multi-modal data including acoustics, vibration, and temperature to obtain comprehensive fault information. This fusion strategy synthesizes heterogeneous sensor data to uncover intrinsic correlations among fault features, thereby improving diagnosis accuracy and system reliability. Sun et al. (2023b) developed a multimodal non-contact diagnosis framework that integrates infrared thermal imaging and acoustic sensing data to form complementary fault information. Specifically, the thermal image was first decomposed into three channels of RGB and the sound signal was transformed into a time-frequency spectrum. Then, cross modal fusion was implemented at the data level, and adaptive weighting of fault sensitive information in the fused features was achieved using a limited shuffling attention module. Li et al. (2024c) proposed a cross-modal zero-shot fault diagnosis framework for rotating machinery that integrates infrared thermography with acoustic data to achieve heterogeneous information complementarity. Within this framework, feature-level fusion was executed at both global and local scales through a composite center fusion module, enabling the construction of holistic fault features. Additionally, a self-enhanced fusion triplet loss function was

incorporated to adaptively reinforce intra-class feature clustering, consequently improving model robustness and generalization capability under varying operational loads.

4. Challenges and future directions

This paper systematically reviews recent advances in non-contact fault diagnosis, categorizing existing approaches into seven major groups: frequency-domain analysis, time-frequency analysis, filtering techniques, statistical feature learning, attention mechanisms, transfer learning, and feature fusion. Through a comparative analysis of these methodologies, several critical issues and promising directions for future research are identified.

4.1. Physics-informed data-driven methods

Despite the remarkable success of data-driven non-contact fault diagnosis methods in industrial applications, there are still some limitations. First, these approaches typically rely heavily on extensive historical datasets. However, in practical engineering scenarios, particularly during the initial operational phases of equipment or under conditions of scarce fault samples, acquiring sufficient high-quality data remains a formidable challenge. Second, data-driven models often lack deep understanding of equipment physics and underlying fault mechanisms, which constrains their generalization capabilities when confronted with unknown or rare failure modes. Furthermore, when processing high-dimensional monitoring data, these methods are susceptible to the “curse of dimensionality,” which can significantly compromise both model performance and interpretability.

To address these challenges, non-contact fault diagnosis methods that integrate data-driven and physics-based modeling have shown promising research prospects. Since physics-based modeling is capable of deeply revealing the intrinsic operating mechanisms of equipment, integrating it with data-driven methods can fully exploit their complementary advantages, thereby simultaneously enhancing diagnostic accuracy and interpretability. Specifically, physics-based models can not only augment training sets by generating synthetic data to improve the generalization capability of data-driven models, but also serve as prior knowledge embedded into the learning process, effectively reducing dependence on large-scale labeled data. Future research may further explore efficient fusion strategies between physics-based modeling and data-driven methods, such as developing novel hybrid model architectures and designing targeted parameter optimization algorithms, to achieve a deeper level of synergy.

4.2. Multi-sensor fusions

Non-contact fault diagnosis methods based on a single sensor predominantly rely on a single data source (Shuai et al., 2023; Yuan et al., 2024) for fault diagnosis. This reliance introduces several inherent limitations, including the lack of information diversity and heightened susceptibility to noise interference, which collectively compromise the accuracy and reliability of fault diagnosis. Moreover, such single-sensor configurations often struggle to comprehensively and accurately identify and localize faults when confronted with complex, evolving fault patterns, resulting in a high risk of missed detections and false alarms. Additionally, the diagnosis capability of a single sensor is fundamentally constrained by its intrinsic performance and precision. Consequently, any degradation or malfunction of the sensor itself can severely impair the effectiveness of the entire fault diagnosis system.

To overcome these limitations, multi-source sensor-based fault diagnosis methods have become increasingly essential. Future research directions can be pursued from the following aspects. First, research on multi-source sensor data fusion technologies should be strengthened to improve fault diagnosis accuracy and robustness by integrating data from different sensor types and positions. For example, advanced data

fusion algorithms such as Kalman filtering and particle filtering can be employed to effectively integrate multi-source data, thereby achieving comprehensive capture and accurate identification of fault features. Secondly, a multi-source fault diagnosis framework based on deep learning is constructed, utilizing attention mechanisms, causal inference, and other techniques to automatically learn the evolution of faults from historical monitoring data. An end-to-end intelligent diagnosis model is established to achieve adaptive recognition and classification of complex fault patterns.

4.3. Integrating expert knowledge with deep learning

In non-contact fault diagnosis based on deep learning, the lack of expert knowledge and prior information can easily lead the model to learn false features unrelated to the fault, significantly weakening its generalization ability under different conditions and fault types, making it difficult to meet practical application needs. For example, in the fault diagnosis of mechanical equipment, the model may ignore specific frequency components in the vibration signal that are closely related to equipment damage, and instead focus on irrelevant noise, resulting in a decrease in diagnosis accuracy. In addition, the “black box” nature of deep learning models further exacerbates the problem of insufficient interpretability. In scenarios such as power equipment fault diagnosis, engineers need to identify the root cause of the fault in order to develop effective maintenance strategies. However, the opacity of the decision-making mechanism within the model hinders this process, increasing the risk and uncertainty of practical applications.

To address these issues, future research should place greater emphasis on integrating expert knowledge and prior information into the design and training of deep learning models. Expert knowledge plays a critical role in this process, helping models gain a deeper understanding of fault mechanisms and enabling the design of more effective fault identification strategies. For example, based on their understanding of fault mechanisms, experts can design specific network architectures or loss functions to strengthen the model’s learning of key features. Taking the practical application of bearing fault diagnosis as an example, engineers have discovered, through studying bearing wear mechanisms, that vibration signals in specific frequency bands hold significant diagnostic value during the bearing degradation process. Leveraging this knowledge, experts can guide deep learning models to focus specifically on these frequency ranges, thereby significantly improving diagnostic accuracy. Furthermore, expert knowledge plays a critical role in feature engineering and data preprocessing. In the signal processing stage, experts can design targeted feature extraction strategies based on prior knowledge of failure modes. Taking gear fault diagnosis as an example, experts can construct specific time-domain or frequency-domain features according to the gear transmission mechanism, guiding the deep learning model to focus on the most discriminative signal representations while suppressing noise and irrelevant interference.

On the other hand, introducing prior information during the training phase helps improve model generalization performance and the ability to recognize rare faults. For example, transfer learning can transfer knowledge from related domains to the target diagnosis task, while semi-supervised learning effectively alleviates the issue of label scarcity by integrating a small amount of labeled data with a large amount of unlabeled data. Furthermore, combining expert knowledge, prior information, and deep learning techniques not only enhances diagnosis accuracy for common faults but also provides more reliable diagnosis results when facing novel or rare faults. This fusion paradigm simultaneously strengthens model interpretability and robustness, offering a solid guarantee for the safe and stable operation of industrial systems.

4.4. Large language models for fault diagnosis

With the rapid development of large language model technologies, their application in the field of fault diagnosis shows great potential, yet

also faces serious challenges. Although large language models can process massive amounts of data and learn complex fault feature patterns, over-reliance on statistical regularities without the guidance of domain knowledge leads to insufficient reliability when transferring across scenarios. To fully unleash the potential of large language models, future research should achieve breakthroughs in the following three directions.

First, it is necessary to optimize the model architecture and training algorithm. By constructing domain knowledge hubs such as a semantic attribute library of equipment faults and a diagnostic rule knowledge graph, and deeply coupling them with large language model architectures, expert knowledge including equipment fault mechanisms. Cross-system fault analogy rules can be embedded into the learning process, significantly enhancing the physical consistency and interpretability of the models.

Second, research should focus on exploring the ability of large language models to intelligently fuse and extract features from multi-source heterogeneous data. This includes multimodal sensor data such as vibration signals, infrared images, acoustic emission, as well as textual information like equipment maintenance logs. Through knowledge-guided data fusion strategies, fault features can be represented more comprehensively. Moreover, to meet the demands of real-time monitoring, lightweight large language model architectures for online diagnosis should be developed, integrating real-time retrieval and reasoning capabilities from domain knowledge bases to satisfy the stringent requirements of industrial scenarios for fast response and continuous diagnosis.

Third, a cross-domain knowledge transfer framework should be constructed. Through methods such as fault mode analogy reasoning and diagnostic rule transfer, the expert knowledge system can be seamlessly integrated with the multi-source data processing capabilities of large-scale models, thereby enabling comprehensive identification of complex fault patterns. This “data-driven plus knowledge-guided” fusion paradigm not only improves the physical credibility of diagnosis results but also drives the evolution of fault diagnosis technology from traditional “black-box” models toward interpretable and transferable intelligent diagnosis systems, bringing breakthrough progress to the reliability management of industrial equipment.

Fourth, continual learning and online adaptive evolution mechanisms should be developed to address the temporal dynamics of industrial equipment. Current large language models are predominantly trained offline with static datasets, whereas machinery exhibits time-varying behavior due to degradation and individualized wear patterns. Future research should investigate parameter-efficient fine-tuning strategies such as Low-Rank Adaptation (LoRA) to enable incremental domain adaptation to new operating conditions or unseen equipment types without catastrophic forgetting or costly full-model retraining. Additionally, online self-supervised updating mechanisms leveraging continuously generated unlabeled field data through contrastive or masked reconstruction tasks can dynamically refresh feature representations, enhancing sensitivity to gradual degradation and incipient anomalies. On the other hand, a human-in-the-loop feedback paradigm should be established, wherein expert validation of diagnostic decisions serves as reinforcement signals to refine decision boundaries and calibrate confidence estimation. This direction bridges the gap between laboratory-trained models and long-term field applicability, ensuring that large language models evolve into self-improving diagnostic agents that maintain reliability throughout the entire operational lifecycle of industrial assets.

5. Conclusion

Non-contact fault diagnosis technology provides innovative solutions for equipment monitoring under extreme operating conditions by eliminating deployment constraints associated with physical contact. This review systematically summarizes the research progress in non-contact fault diagnosis. At the sensing level, non-contact sensors including acoustic sensors, infrared thermal imagers, and laser Doppler vibrometers each possess distinct advantages for different application

scenarios. At the methodological level, intelligent diagnostic capabilities have been continuously enhanced through the evolution from traditional time-frequency analysis and signal decomposition to deep learning frameworks incorporating attention mechanisms, transfer learning, and feature fusion strategies.

Current methods still face challenges including low SNR, poor cross-domain generalization, feature redundancy, and insufficient model interpretability. Future research should focus on four transformative directions. First, physics-informed data-driven integration to enhance model credibility. Second, multi-sensor fusion to achieve information complementarity. Third, expert knowledge embedding to improve interpretability. Fourth, large language model adaptation to expand the frontier of diagnostic intelligence. This article will propel non-contact fault diagnosis toward knowledge-guided, trustworthy, and reliable next-generation intelligent operation and maintenance systems.

CRedit authorship contribution statement

Dan Zhang: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation; **Yongyi Chen:** Writing – review & editing, Data curation; **Zhong Liu:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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